One-variable factorization and approximate additive linearization

Author proposal: operator-local agent /root/astra_v14_baseline_compiler

Protocol 1.4; exact source contracts for the Sections 9–11 repair

This is a new connecting proof, pending independent review. It does not approve itself, the complete inverse theorem, or any release. The earlier Section 13 packet is unchanged. We use the final published sources [GTZ], [GT12], [GT14], and [GT08] identified in the accompanying source record. Only one-variable factorization is asserted here.

1 Objects, source differences, and the required outputs

All nilpotent Lie groups below are standard, connected and simply connected; all lattices and filtrations are standard and rational. Subgroups described as rational are connected, and their intersection with the ambient lattice is cocompact in that subgroup. We use the natural ultrapowers of these objects, with the internal-function convention of [GTZ, Appendix A]. A bounded quantity is bounded by some standard constant. In the finite combinatorial proof all constants depend only on the stated positive density and error constants. Passing that finite proof to an ultrapower therefore preserves uniformity and internal choices. Braces denote the representative in $(-1/2, 1/2]$.

An $\mathbb{N}$-filtration $G_\bullet$ may have $G_1 \neq G_0 = G$. Its positive groups satisfy $[G_i, G_j] \subseteq G_{i+j}$ and are rational. The polynomial convention is the ordinary degree filtration on the source. In fixed adapted Mal'cev coordinates, write the coordinates of a polynomial map as a finite polynomial in n or $\binom{n}{j}$. Coordinate degree is bounded in terms of the fixed filtered group. For a smooth factor, "small Taylor coefficients" means that its constant coefficient is bounded and its coefficient of index $j \geq 1$ is $O(N^{-j})$, in the group Taylor convention of [GTZ, Lemma B.9]. Equivalent coordinate bounds are justified below.

The statement of [GTZ, D.2] cannot be used as a literal equivalence: on $\mathbb{R}/\mathbb{Z}$, $g(n) = n/N$ is equidistributed on $[N]$ but is not equidistributed on $[\lfloor N/2 \rfloor]$. For example its first Fourier average on that half interval tends to $2i/\pi$, whereas its full-interval average tends to zero. Its nontrivial horizontal character is $1/N$-Lipschitz. We instead use the one-variable weak factorization [GT12, Proposition 9.2], whose proof uses the quantitative theorem on an actual dense progression. The published correction [GT14, pp.1175–1176] explicitly retains the one-variable results. Its minor corrections on pp.1182–1183 are part of this source choice.

The properness requirement printed in the second bullet of [GTZ, D.4] is removed. Only a weak-factorization step that starts with failed total equidistribution forces a proper subgroup. Also, a rational periodic factor is not asserted to take values in Γ.

Theorem 1 (One-variable factorization, DF-D4)**.** *Let N be unbounded and let $g \in {}^*\mathrm{poly}(\mathbb{Z}, G_\bullet)$. There are a standard connected rational subgroup $P \leq G$ and internal polynomial maps $\epsilon, p, \gamma \in {}^*\mathrm{poly}(\mathbb{Z}, G_\bullet)$ such that*

$$g(n) = \epsilon(n)p(n)\gamma(n) \qquad (n \in {}^*\mathbb{Z}).$$

The map p takes values in ${}^*P$, with its induced filtration. On $[N]$, ϵ is bounded, is $O(1/N)$-Lipschitz, and has the small Taylor coefficient bounds above. There is a standard discrete subgroup $\Lambda \geq \Gamma$

with $\gamma(n) \in {}^*\Lambda$ for every n, and a standard integer $q > 0$ with $\gamma(n+q)\Gamma = \gamma(n)\Gamma$ for every n. For every standard finite-index subgroup $\Delta \leq P \cap \Gamma$, $p(n)\Delta$ is totally equidistributed in P/Δ on $[N]$. The subgroup P is allowed to equal G or to be trivial.

Theorem 2 (Approximate four-function linearization, DF-F1)**.** *Let N be unbounded, let $j \geq 1$ be standard, and put $e = N^{-j}$. Let $H \subseteq [-N, N] \cap {}^*\mathbb{Z}$ be internal, and let $\xi_i : H \to {}^*\mathbb{T}$, $1 \leq i \leq 4$, be internal functions. Suppose that for at least cN^3 ordered quadruples in H^4 satisfying $h_1 + h_2 = h_3 + h_4$ one has*

$$\|\xi_1(h_1) + \xi_2(h_2) + \xi_3(h_3) + \xi_4(h_4) - a\|_{\mathbb{T}} \leq Ce,$$

where $c > 0$ and $C < \infty$ are standard and $a \in {}^*\mathbb{T}$. For any specified index i there are standard $K < \infty$, $c' > 0$, an internal $H' \subseteq H$ with $|H'| \geq c'N$, internal $\alpha_k \in {}^*\mathbb{T}$, $\beta_k \in {}^*\mathbb{R}$, and $\delta \in {}^*\mathbb{T}$ such that

$$\xi_i(h) = \delta + \sum_{k=1}^{K} \{\alpha_k h\}\beta_k + O(e) \quad (h \in H'). \tag{F}$$

The implicit constant is standard and uniform over H'. The bracket sum restricted to H' is an exact Freiman homomorphism: it preserves every relation $h_1 + h_2 = h_3 + h_4$ with all four points in H'. No initial Freiman property, boundedness of the real β_k, or regularity of the input functions is assumed. All choices may be made internally.

2 Weak factorization and coefficient bounds

Lemma 3 (Weak step, DF-D3)**.** *Suppose first that $G_1 = G$. If $g(n)\Gamma$ is not totally equidistributed on $[N]$, it admits the factorization of Theorem 1 with p taking values in a standard proper connected rational subgroup, without the final equidistribution requirement. All other bounds and the discrete-overgroup and period assertions hold.*

Proof. Failed total equidistribution supplies an internal dense progression and a standard Lipschitz test with noninfinitesimal discrepancy. Thus for some standard $0 < \delta < 1/2$ it fails total δ-equidistribution in the quantitative convention of [GT12]. Reduce δ further to include the rationality constant of the fixed Mal'cev basis. Apply the one-variable case of [GT12, Proposition 9.2, pp.522–523] with this δ, the actual degree of $G_\bullet$, and the actual dimension of G. Its bounds are standard. The same argument works for an internal Lipschitz test with a standard norm bound: a finite uniform net in the standard Lipschitz ball first supplies a standard test with positive discrepancy. This uses compactness of G/Γ.

Here are the stronger bounds extracted from that source's construction, including the distinctions needed for later use. Write $g(0) = c\lambda$ with c in a fixed bounded fundamental set and $\lambda \in {}^*\Gamma$. Then $\widetilde{g} = c^{-1}g\lambda^{-1}$ has value 1 at zero. Left translation by bounded c changes a Lipschitz norm by a standard factor, and right multiplication by λ^{-1} changes no point of G/Γ. Thus the same failed-total-equidistribution argument applies to $\widetilde{g}$, with a possibly smaller standard δ.

The source's horizontal-character step gives a nonzero standard integer vector k with

$$\psi(\widetilde{g}(n)) = \sum_{r \geq 1} t_r \binom{n}{r}, \qquad \|k \cdot t_r\|_{\mathbb{T}} = O(N^{-r}). \tag{D1}$$

Here $k \cdot \psi$ is the real lift of the horizontal homomorphism; in particular it kills all commutators. Standardness of k follows because the quantitative bound leaves finitely many integer vectors. The source obtains (D1) by applying its Theorem 2.9 to the witnessing progression and undoing that

progression by its Lemma 8.4. This use does not assert the false single-scale equivalence described above.

Choose integers a_r nearest to $k \cdot t_r$. The coordinate support condition in [GT12, Lemma 6.7] says that t_r lies in the coordinate subspace for G_r. If k vanishes on that subspace, take $a_r = 0$ and $v_r = w_r = 0$. Otherwise fix a rational vector u_r in that subspace with $k \cdot u_r = 1$, and put

$$v_r = a_r u_r, \qquad w_r = (k \cdot t_r - a_r)u_r.$$

Define ϵ_0 and γ_0 by Mal'cev coordinates

$$\psi(\epsilon_0(n)) = \sum_r w_r \binom{n}{r}, \qquad \psi(\gamma_0(n)) = \sum_r v_r \binom{n}{r}, \qquad p = \epsilon_0^{-1}\widetilde{g}\gamma_0^{-1}. \tag{D2}$$

The same coordinate-support lemma and closure of polynomial maps under products show that these are polynomial for the original filtration. The real horizontal lift vanishes identically on p, since its coefficient at every index is $k\cdot t_r-(k\cdot t_r-a_r)-a_r = 0$. Consequently p lies in the connected kernel of that real homomorphism. This is a standard proper rational subgroup. A kernel of the circle-valued homomorphism alone would not establish connectedness; (D2) supplies the real zero required here.

Each w_r is $O(N^{-r})$. The vectors v_r have a common standard denominator, but their numerators need not be bounded. This denominator statement is all that is required. By [GT12, A.11–A.12], γ_0 is uniformly rational and periodic modulo Γ with a standard period. The more explicit common discrete overgroup argument below also applies. Set $\epsilon = c\epsilon_0$ and $\gamma = \gamma_0\lambda$. These are the correct ordered factors of the original g. Multiplication by bounded c preserves the stated small-coordinate bounds, and the added λ lies in the original lattice. The next lemma verifies all closure and Taylor assertions used here and in the descent. □

Lemma 4 (Closure of the controlled factors)**.** *In a fixed rational nilpotent group, finite products of bounded polynomial maps with coefficient bounds $O(N^{-r})$ retain those bounds and are $O(1/N)$-Lipschitz on $[N]$. These bounds imply the group Taylor bounds in Theorem 1. A finite collection of rational lattices commensurable with Γ, together with sets of points whose rational coordinates have a fixed common denominator, is contained in a standard discrete subgroup $\Lambda \geq \Gamma$. Any fixed-degree polynomial map into this group is periodic modulo Γ with a standard period.*

Proof. Coordinate products, inverses, and changes between first-kind and Mal'cev coordinates are fixed polynomials by [GT12, A.2–A.3]. In a coordinate polynomial $\sum a_r n^r$, the bounds $a_r = O(N^{-r})$ are equivalent to bounded coefficients after substituting $n = Nt$. Products of such polynomials retain that property. Converting back to the binomial basis gives at index r a fixed linear combination of coefficients of monomial indices $\ell \geq r$, hence $O(N^{-r})$. Constant coefficients stay bounded. On the fixed compact coordinate range for $0 \leq t \leq 1$, the group-coordinate maps have bounded derivatives; differentiation in n contributes N^{-1}. This proves the Lipschitz assertion, with any fixed standard metric.

For clarity about the group Taylor convention, one may assign formal weight r to each coefficient of n^r in the logarithm of a coordinate polynomial. The recursive Taylor extraction by $g(0)$, then $g_1^{\binom{n}{1}}$, then $g_2^{\binom{n}{2}}$, and so on uses only fixed BCH polynomials. Its coefficient $\log g_r$ contains no monomial of formal weight less than r. Indeed after removing indices below r, each surviving polynomial term of formal weight w has ordinary degree at most w. The residual group map is the identity at $0, \ldots, r-1$, as a formal identity in its input coefficients. Its logarithmic terms of weight $w < r$ are therefore polynomials of degree less than r with r zeros, and vanish. BCH

products add weights, so this assertion is preserved at each extraction step. The remaining finitely many monomials have weight at least r, and are $O(N^{-r})$. This proves the group Taylor assertion, including under products and fixed rational embeddings of subgroups.

For the discrete-overgroup assertion, use a rational basis adapted to the lower central series and let its weights be $w_i \geq 1$. All points of each given lattice have logarithmic coordinates with bounded denominators, as do the specified coordinate sets, by the fixed polynomial coordinate changes. Choose a standard integer D divisible by these denominators and by the denominators of all coefficients in the finite BCH formula in this basis. Increase D for the finitely many basis conversions if necessary. The set

$$L = \sum_i D^{-(2w_i-1)}\mathbb{Z}X_i$$

is closed under BCH. For a nonlinear term with $t \geq 2$ inputs of weights $w_{i_1}, \ldots, w_{i_t}$ contributing to output weight w_k, one has $w_k \geq \sum w_{i_a}$. Therefore the difference of denominator exponents is

$$(2w_k - 1) - \sum_a (2w_{i_a} - 1) \geq t - 1 \geq 1.$$

The extra power of D clears that BCH coefficient. Linear terms and negation also preserve L. Thus $\exp L$ is a standard discrete subgroup containing all the required points and Γ. It is cocompact, and $[\exp L : \Gamma]$ is finite. This argument also proves commensurability claims needed when a lattice of a rational subgroup is enlarged inside G.

For the period in the required right-coset convention, write the Mal'cev coordinate vector of $\gamma(n)$ as a polynomial of fixed degree D_0. Its values have a common standard coordinate denominator, since $\gamma(n) \in {}^*\Lambda$. Interpolating at $0, \ldots, D_0$ shows that its coefficients have a common standard denominator too. Their numerators may be arbitrary internal integers. By the rational polynomial formulas for multiplication and inversion, the vector

$$R(n, r) = \psi\big(\gamma(n)^{-1}\gamma(n + r)\big)$$

is a polynomial in the two integer variables n, r with a common standard coefficient denominator A. The identity $R(n, 0) = 0$ is a polynomial identity, so every monomial has a positive power of r. For the standard integer $q = A$, every coordinate of $R(n, q)$ is an integer for every internal integer n. Adapted Mal'cev coordinates identify Γ with the integer-coordinate points. Hence $\gamma(n)^{-1}\gamma(n + q) \in {}^*\Gamma$, exactly the assertion $\gamma(n + q)\Gamma = \gamma(n)\Gamma$. This proves the conclusion of [GT12, A.12] directly; its printed left quotient $\gamma(n + q)\gamma(n)^{-1}$ is not the quotient used here. □

3 Minimal subgroup descent, with all finite covers retained

Proof of Theorem 1. First address the filtration convention. If G_1 is proper, write $g(0) = c\lambda$ as above and set $p = c^{-1}g\lambda^{-1}$. Then $p(0) = 1$ and

$$p(n) = \lambda\big(g(0)^{-1}g(n)\big)\lambda^{-1} \in G_1.$$

The groups G_r are normal in G_0, so conjugation preserves them. The induced filtration on G_1 has zeroth and first group both G_1, and the positive groups remain G_r. Hence the one-variable result with the convention $G_0 = G_1$ applies there. This also shows directly that a controlled representation exists: $\epsilon = c$, $\gamma = \lambda$, and the represented subgroup is G_1. If $G_1 = G$, the identity-factor representation is already enough. Trivial groups cause no exception.

Call a subgroup P a representing subgroup if a factorization with all conclusions of Theorem 1 except total equidistribution exists. Choose one with least dimension among the nonempty collection of standard rational representing subgroups. This uses a least element of the finite set $\{0, \dots, \dim G\}$, not an infinite iteration or a uniform choice over all finite covers.

Suppose that for a standard finite-index $\Delta \le P \cap \Gamma$ the orbit $p\Delta$ is not totally equidistributed. The induced filtration of P is rational for Δ, since passing to a finite-index lattice does not change its rational Lie algebra. If $P \cap G_1$ is proper in P, the normalization just given already represents p in that proper subgroup, with a bounded constant on the left and a Δ-element on the right. Otherwise apply Lemma 3 in P/Δ to obtain

$$p = \epsilon_2 p_2 \gamma_2$$

with p_2 in a standard proper rational $P_2 < P$. Rationality relative to Δ is rationality relative to $P \cap \Gamma$, so P_2 is rational in G. The actual new factorization is

$$g = (\epsilon\epsilon_2) p_2 (\gamma_2 \gamma). \tag{D3}$$

The order of both products is necessary in a noncommutative group. Lemma 4 proves the small-coefficient assertions for the left product, in the ambient coordinates. For the right product, the rational lattice coordinates of γ_2 relative to Δ have common standard denominators in the original rational structure. The same lemma supplies a single discrete overgroup containing its values, the old overgroup, and Γ. It also supplies the standard period modulo Γ. Thus (D3) is a representation by the lower-dimensional subgroup P_2, contrary to minimality.

No choice of Δ was imposed in that minimality argument. Consequently the chosen p is totally equidistributed for every standard finite-index $\Delta \le P \cap \Gamma$, with the same factors. This is the required quantifier order. The argument only produces standard constants for this particular internal orbit and fixed standard geometry. It does not assert a prescribed effective bound uniform over all internal orbits. □

4 A finite combinatorial proof of the linearization contract

We prove a finite assertion with constants depending only on c, C and then transfer. Take $N \ge 1$, $0 < e \le 1$, and suppose the hypotheses of Theorem 2 hold in finite sets. Put $M = \lceil 1/e \rceil$ and $Q = 100N+1$. Round each $\xi_i(h)$ in the circle to $y_i(h) \in M^{-1}\mathbb{Z}/\mathbb{Z}$, with error at most $1/(2M) \le e/2$. Since $Me \le 2$, the approximate sum can take only a bounded number, depending on C, of values in this grid. For at least $c_1 N^3$ of the original quadruples its rounded value is a fixed a_0 in that grid. Here $c_1 > 0$ depends only on c, C.

Embed $[-N, N]$ in $B = \mathbb{Z}/Q\mathbb{Z}$ and identify the target grid with $A = \mathbb{Z}/M\mathbb{Z}$. In $B \times A$ form the four sets

$$A_1 = \{(h, y_1(h))\}, \quad A_2 = \{(h, y_2(h))\}, \quad A_3 = \{(-h, y_3(h))\}, \quad A_4 = \{(-h, y_4(h))\}. \tag{F1}$$

The signed first coordinates are essential. The selected quadruples now give $c_1 N^3$ representations of $(0, a_0)$ in $A_1 + A_2 + A_3 + A_4$. The initial additive relation has no wraparound, since $4N < Q$. Fourier inversion and Holder's inequality in this finite abelian group give

$$r_{A_1+A_2+A_3+A_4}(0, a_0) \le \prod_{i=1}^{4} E(A_i)^{1/4}, \qquad E(A_i) = |\{x_1 + x_2 = x_3 + x_4 : x_a \in A_i\}|.$$

For completeness, with unnormalized Fourier transforms the inverse formula has factor $|B \times A|^{-1}$; Holder bounds the sum of four transforms by the fourth moments, each of which equals $|B \times A|E(A_i)$. Those factors cancel. Each $E(A_i) \le |H|^3 \le 27N^3$. It follows that $E(A_1) \ge c_2 N^3$ for a positive $c_2 = c_2(c, C)$. The same reasoning can isolate any A_i; reflecting its first coordinate back if necessary gives the graph of the specified y_i. We henceforth call that function y and its graph A_1.

Apply [GT08, Theorems 5.2 and 5.3, p.95] to A_1 in the actual finite group $B \times A$. Their hypotheses follow from the energy bound and $|A_1| \le 3N$. They give a subset Γ' of its graph and a constant $K_0 = K_0(c, C)$ such that

$$|\Gamma'| \ge N/K_0, \qquad |9\Gamma' - 8\Gamma'| \le K_0 N. \tag{F2}$$

We use these published theorems themselves, not the special correlation premise in [GT08, Proposition 5.4].

Graph refinement with the actual cyclic target

Let $V = \{a \in A : (0, a) \in 8\Gamma' - 8\Gamma'\}$. The map $\Gamma' \times V \to 9\Gamma' - 8\Gamma'$ given by addition is injective: equality determines the first graph coordinate, then its unique second coordinate, and finally a. Hence $|V| \le K_0^2$. There is a set S_A of at most $O(1 + \log K_0)$ characters of A for which

$$V \cap \{a : \|\chi(a)\|_{\mathbb{T}} < 1/4 \ \forall \chi \in S_A\} = \{0\}.$$

Indeed, for any nonzero a a uniformly random character takes values uniformly among the roots of unity of its order; at least one third of those values have distance at least $1/4$ from zero. Choose enough independent characters that the union bound over $V \setminus \{0\}$ is less than one. This proves the required separation without identifying A with the dual of B, an identification that is unavailable here.

Partition each coordinate of the character vector on A into intervals of length at most $1/100$. A cell contains a subset $\Gamma'' \subseteq \Gamma'$ of size $c_3 N$. In any element of $8\Gamma'' - 8\Gamma''$, every selected character of the second coordinate has a real representative of absolute value less than $16/100 < 1/4$ when its first coordinate is zero. It follows that the only element of this difference set with first coordinate zero is $(0, 0)$. Therefore $4\Gamma'' - 4\Gamma''$ is a graph. Write H'' for the projection of Γ'', and let

$$\zeta : 2H'' - 2H'' \longrightarrow A$$

be the function whose graph is $2\Gamma'' - 2\Gamma''$. It is an exact Freiman homomorphism on this domain: two sums of two of its graph points lie in $4\Gamma'' - 4\Gamma''$, where the first coordinate determines the second. Also $\zeta(0) = 0$. Thus $\zeta(x + y) = \zeta(x) + \zeta(y)$ whenever $x, y, x + y$ lie in its domain. This proves precisely the generalization of [GT08, Lemma 9.2] needed here.

A proper progression and an explicit linear bracket formula

The set H'' has density at least $c_3/101$ in B. By [GT08, Lemma 6.4, pp.97–98] there is a bounded set of characters S with $B(S, 1/4) \subseteq 2H'' - 2H''$. Add the faithful character $x \mapsto x/Q$ to S, and shrink the radius to $\rho = 1/16$. Write $d = |S|$, a bounded integer. [GT08, Lemma 10.4, p.121] supplies a proper centered progression

$$P = \Big\{ \sum_{a=1}^{r} l_a v_a : |l_a| < L_a \Big\} \subseteq B(S, \rho), \qquad |P| \ge c_4 Q,$$

with $r \le d$, and linearly independent vectors $\Phi(v_a) = (\{\chi(v_a)\})_{\chi\in S}$. The coset subgroup in that lemma is zero because S contains a faithful character. Its proof also gives

$$\Phi\left(\sum_a l_a v_a\right) = \sum_a l_a \Phi(v_a) \quad \text{on } P. \tag{F3}$$

This is the lift of a lattice progression inside the real cube, not an unjustified inference from reduction modulo one. Delete inactive generators with no nonzero integer coefficient allowed; all remaining v_a lie in P. Every partial sum along a coordinate path to a point of this centered box stays in P. The local additivity of ζ consequently gives

$$\zeta\left(\sum_a l_a v_a\right) = \sum_a l_a \zeta(v_a).$$

Choose real lifts b_a of $\zeta(v_a)$ and real dual vectors u_a to the independent vectors $\Phi(v_a)$ on their span. Using (F3), define $b_\chi = \sum_a b_a (u_a)_\chi$. Then, for every $p \in P$,

$$\zeta(p) = \sum_{\chi\in S} b_\chi \{\chi(p)\} \pmod 1. \tag{F4}$$

This is a linear calculation. It uses neither the quadratic-extension theorem nor the quadratic classification lemma in [GT08]. The author notice concerning that extension theorem is therefore not a new assumption here.

Translation, fixed carries, and the Freiman restriction

Averaging translates of P in B gives an $x \in B$ with $|H'' \cap (x + P)| \ge |H''||P|/Q \ge c_5 N$. For $h = x + p$ and $h_0 = x + p_0$ in this intersection, the graph relation gives $y(h) - y(h_0) = \zeta(p - p_0)$, since $\Gamma'' - \Gamma''$ is contained in $2\Gamma'' - 2\Gamma''$. Both p, p_0 lie in the domain of ζ. Its Freiman identity applied to $(p - p_0) + p_0 = p + 0$ gives

$$y(h) = \delta_x + \zeta(p).$$

For each $\chi \in S$, write $\chi(h) = \alpha_\chi h$ on the cyclic group. The difference

$$\{\chi(h - x)\} - \{\chi(h)\} + \{\chi(x)\}$$

is an integer in a fixed set of at most three values. Restrict to a cell on which all these carries are fixed. This costs a factor at most 3^d and, by (F4), gives

$$y(h) = \delta + \sum_{\chi\in S} b_\chi \{\alpha_\chi h\}.$$

Finally partition each $\{\alpha_\chi h\}$ into a fixed number of intervals of length at most $1/10$, and retain a largest cell H'. For any integer relation $h_1 + h_2 = h_3 + h_4$ in H', the corresponding sum of four fractional parts, with signs $+, +, -, -$, is an integer of absolute value at most $1/5 < 1$. It is therefore zero in $\mathbb{R}$ for each character. The real weighted sum is an exact Freiman homomorphism on H'. Undoing the rounding changes the displayed value by at most $e/2$. All density losses depend only on c, C and all sets and choices are finite. This proves the finite assertion, and transfer with $e = N^{-j}$ proves Theorem 2. A bounded rank can be made a fixed standard rank by passing to its ultrapower value or by adding zero terms.

5 Consumer boundary and related-use scope

Theorem 1 supplies exactly the one-variable factorization consumed by the Sections 9–11 repair. That repair still must derive its nonzero coset integral and its degree-rank horizontal inclusion for the actual Taylor quotients. In particular, it may multiply a character by a standard denominator to kill the rational periodic contribution; it may not silently erase that contribution. No such application is accepted by this packet.

Theorem 2 applies separately at each of the finitely many required indices j, including $j = 1$, and to any chosen one of the four functions. The number of applications is standard. Dense refinements and all losses remain standard only when that finite iteration is recorded by the consumer. The theorem supplies no global Freiman extension beyond H'.

The related-use audit is bounded by uses of the single-scale equivalence, properness in D.4, composition of factorization factors, deletion of rational Taylor contributions, and the graph/Bohr/bracket joins in F.1. The multi-variable D.5 and its consumers in Appendix E require their own exact application review. They are not inferred from the one-variable theorem here. Likewise, this packet does not accept Gaussian's pending Sections 9–11 argument or any downstream assembled inverse theorem.

References

[1] [GTZ] B. Green, T. Tao and T. Ziegler, An inverse theorem for the Gowers $U^{s+1}[N]$-norm, *Annals of Mathematics* 176 (2012), 1231–1372. DOI: 10.4007/annals.2012.176.2.11.

[2] [GT12] B. Green and T. Tao, The quantitative behaviour of polynomial orbits on nilmanifolds, *Annals of Mathematics* 175 (2012), 465–540. DOI: 10.4007/annals.2012.175.2.2.

[3] [GT14] B. Green and T. Tao, On the quantitative distribution of polynomial nilsequences—erratum, *Annals of Mathematics* 179 (2014), 1175–1183. DOI: 10.4007/annals.2014.179.3.8.

[4] [GT08] B. Green and T. Tao, An inverse theorem for the Gowers $U^3(G)$ norm, *Proceedings of the Edinburgh Mathematical Society* 51 (2008), 73–153. DOI: 10.1017/S0013091505000325.